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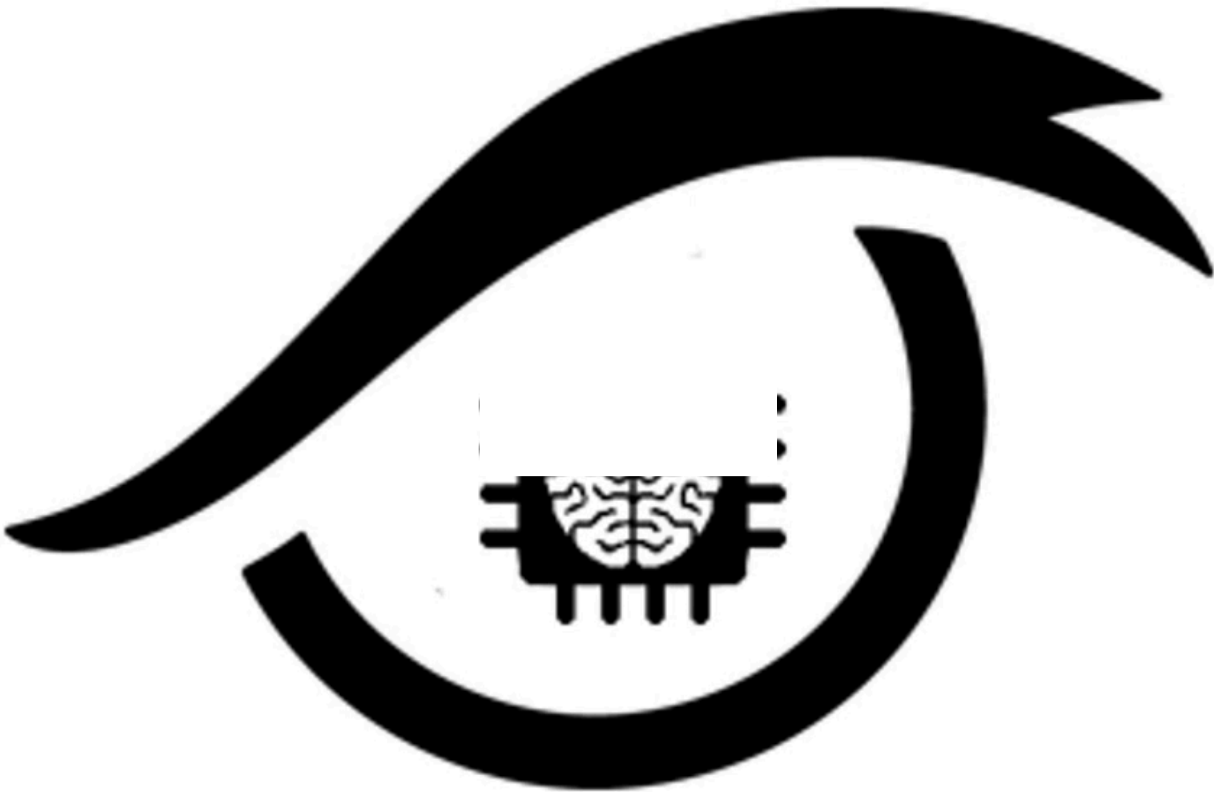
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Fraud Detection and Prevention with AI

Hawkeye Platform: Real-Time AI-driven Fraud Detection and Prevention Solutions



Oleg Romanov

First created • 0 backed

More

\$714



pledged of \$19,065 goal

3

backers

 Software

 Toronto, Canada

 Project budget

Funding Unsuccessful

The project's funding goal was not reached on Thu, April 1 2021 12:19 AM UTC +06:00

 **Last updated** [April 4, 2021](#)[Campaign](#)[Rewards](#)[Creator](#)[FAQ](#)[Updates](#) ¹[Comments](#) ⁰[Save](#)

Story

Why The Fuss

Over the past decade the online share of total worldwide retail sales has steadily been growing, reaching well over \$4 trillion in 2020.

Unfortunately, the other side of the coin is a continuously growing number of fraud attacks - global losses from payment fraud has tripled from \$9.84 billion in 2011 to \$32.39 in 2020, of which the United States claims more than one third.

The unprecedented jump in payment fraud we've seen since the onset of global pandemic is attributed not only to a drastic increase in e-commerce activity, but also to the fact that in all times fraudsters take advantage of confusion, uncertainty and vulnerability caused by COVID-19 crisis. As both businesses and consumers are struggling to stay afloat due to a decline in the global economy, fraudsters are targeting them more aggressively than ever. A recent TransUnion survey found that 22% of Americans have been targeted by fraud scams related to COVID-19. A new study from [Juniper Research](#) forecasts that businesses in eCommerce, airline ticketing, money transfer and banking services, will cumulatively lose over \$200 billion to online payment fraud between 2020 and 2024.

Fraudsters are constantly developing new schemes and using advanced technologies to be at the top of their game. The only way for merchants to combat this is to invest in the right risk management solution.

About Us

HawkEye Risk Solutions is formed by a group of experts in Payments who have successfully worked on fraud prevention for the past 15 years. We decided to join our efforts and use our vast interdisciplinary knowledge of the industry, internet security, and data science to build an effective fraud detection tool that would help businesses grow by protecting them against fraud.

Having a strong mathematical background, we decided to take advantage of Artificial Intelligence and Data Science to fight fraud. We have built a production-ready AI-driven Fraud Detection platform for payment

providers, and now we also have a working prototype of a Real-Time AI-driven Fraud Detection platform for online merchants.

Our Product At A Glance

The HawkEye Platform – payment fraud detection engine, which delivers risk mitigating decisions in real-time. It incorporates multiple AI tools and decision layers to analyze transaction flows in a holistic way and unlock hidden insights in fraudulent behaviors.

Layer 1: HawkEye Rule Engine

The Rules Engine is the first line of defense. It captures a high amount of fraud by screening transactional data against a set of rules derived from known fraud patterns. Our rules monitor various dimensions pertaining to payments, such as payment amount, frequency of declined authorization attempts, pattern of duplicated card numbers, number of refunds, etc. Using them we detect suspicious transactions.

Layer 2: HawkEye Supervised Machine Learning Model

In the second layer, we use a fully connected Deep Neural Network (DNN) to classify a transaction as fraud or not fraud. To do this, we first use merchant's historical data – where each transaction is labeled as either fraud or non-fraud – to train a supervised learning model to classify transactions. Then, we apply this model on top of the existing rules engine to distinguish the fraudulent transactions that may have not been alerted by the rules.

Layer 3: HawkEye Autoencoder

Here we attempt to uncover some cases of fraud that got past the Rules Engine and Supervised Model. Our Autoencoder is trained to create feature representation of a valid transaction so it can effectively detect deviations from the norm, even if such deviations are very subtle and only noticeable when considering a series of events and seemingly uncorrelated features. If a deviation is above some pre-set threshold it is considered a candidate for fraud.

Clients can connect to HawkEye Platform using REST API so the decisions can be easily integrated into their information systems.

Why Us?

1. We are an independent data science shop operated by technology and

industry experts who have worked in the field for more than a decade.

2. We have a state-of-the-art fraud detection framework that effectively integrates traditional and modern AI-based fraud detection methods.

Utilizing this omnichannel approach increases efficiency and versatility of our platform, thereby making it a core instrument in the battle against online payment fraud.

3. We stay on top of emerging fraud trends and latest technological developments. We continuously monitor, audit, scale, and enhance our platform so it's well-equipped even against most sophisticated fraud attacks.

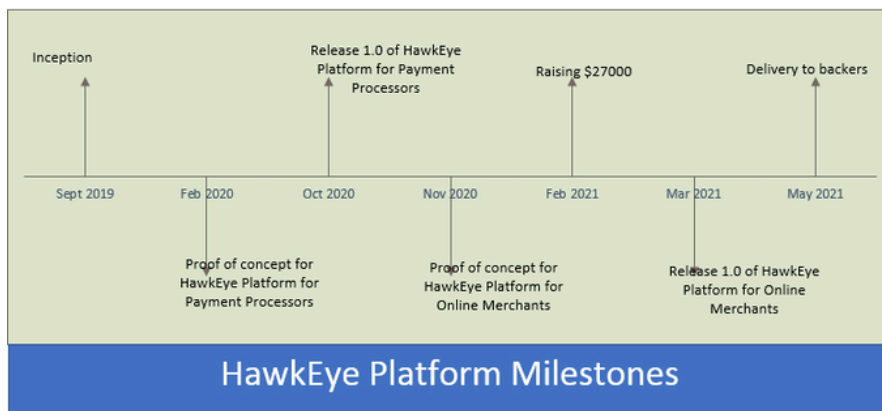
4. Our solution is equally suitable for merchants processing large volume of transactions and those operating on a smaller scale.

5. We offer a 7-day free trial in order for our customers to experience the benefits of our service before having to pay for it.

Ultimately we're convinced that The HawkEye Platform can help business grow by protecting them against fraud.

Our Goal

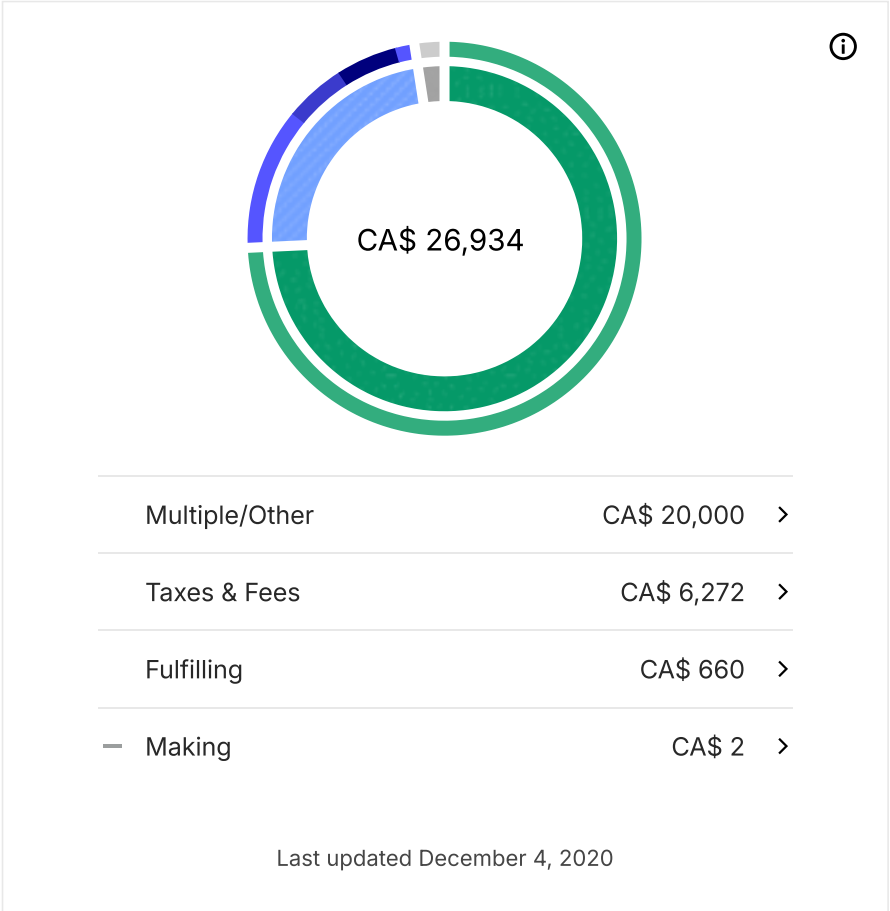
Our goal is to raise 27K during this campaign. If successful, we will need another 2 months to move our HawkEye Platform for Online Merchants from a prototype state to a production scaled solution. We're expecting to delivering a solution to our backers in May - June 2021.



Our cost structure breaks down as the following:

Cost Allocation	Total Cost
Labor	\$20 000,00
Founders, 320 hours x CA\$20.00	\$6 400,00
Software Development, 160 hours x CA\$50.00	\$8 000,00
Marketing & Sales, 160 hours x CA\$35.00	\$5 600,00
Operational Expense	\$660,00
Server Hosting, 1 month, 3 servers x CA\$220.00	\$660,00
Taxes	\$2 685,80
Ontario HST, 13%	\$2 685,80
Fees	\$2 334,58
Kickstarter fee, 5%	\$1 167,29
Payment processing fee, 5%	\$1 167,29
Reserves	\$1 167,29
Reserve for miscalculations, 5%	\$1 167,29
GRAND TOTAL	\$26 847,67

Project budget



This is a projected budget provided by the creator and may be subject to change.

Risks and challenges

Our backers get one-month subscriptions at 50% discount as a reward.

There is a challenge to retain the backers as our clients after the first month at the full pricing.

[Learn about accountability on Kickstarter](#)

Environmental commitments

[Visit our Environmental Resources Center](#) to learn how Kickstarter encourages sustainable practices.

Something else

We will employ our developers and analysts to work from homes reducing our office space and footprint associated with that.

We will choose our office and server hosting locations that are managed by companies that minimize their environmental impact.

Questions about this project? [Check out the FAQ](#)

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 Save our oceans. Take Wildtype mainstream.

Justin Kolbeck



Funding period

Jan 31 2021 - Apr 1 2021 (60 days)



 Zen Tea Brew Infusions: Discover Delicious Rare Shizuoka Tea



 Faire in the Field: Year Two!

Patrick Munroe-Smith



 Laughing Gull Chocolates Take Flight

Lindsay

 10 days left • 81% funded





English ▾

\$ US Dollar (USD) ▾

